

Lab: Feeling the Boundary of Your Skin

Objective

Experience the body as a living system with a boundary through somatic exercises that explore the felt difference between inside and outside, self and world. This lab introduces the Markov blanket as the skin -- your first-person encounter with the concept of a system boundary.

Prerequisites

- A quiet, comfortable space where you can sit or stand undisturbed for 25 minutes
- Optional: a partner for Part 4
- No prior knowledge required -- only willingness to attend to sensation

Part 1: The Skin Envelope (5 minutes)

Stand or sit with your eyes closed. Bring your attention to the surface of your body. Begin at the crown of your head and slowly scan downward -- forehead, cheeks, neck, shoulders, arms, torso, hips, legs, feet.

At each area, notice: Where does "me" end and "not-me" begin? The skin is your boundary. In Active Inference, this is the Markov blanket -- the surface where internal states meet external states.

Notice places where the boundary feels sharp (fingertips, lips) and places where it feels vague (the back, areas under clothing).

Write your response here

(Describe three places where your boundary felt most vivid and three where it felt most diffuse. What made the difference?)

Part 2: Inside and Outside (5 minutes)

Keep your eyes closed. Now alternate your attention:

1. **Inside:** For 30 seconds, attend only to what you feel inside -- heartbeat, breath, warmth, tension, the weight of organs.
2. **Outside:** For 30 seconds, attend only to what reaches you from outside -- sounds, air temperature, the pressure of the chair.
3. Repeat the alternation three times, making the switches faster each time.

In Active Inference terms, you are shifting between interoception (sensing internal states) and exteroception (sensing external states). The boundary between these two streams of information is what makes you a system.

Write your response here

(What happened when you switched quickly between inside and outside? Was the boundary between them clear or blurry?)

Part 3: Perturbing the Boundary (7 minutes)

Now explore what happens when the boundary is disrupted or extended:

1. **Cold contact:** Place something cool (a glass of water, a metal object) against your forearm. Notice how the boundary of your skin becomes intensely vivid at the point of contact. This is prediction error -- the body did not expect this temperature.
2. **Tool extension:** Pick up a pen or stick and tap the table surface. Notice: after a few taps, do you feel the table through the tip of the tool? Your body boundary has extended. In 4E cognition terms, the tool has become part of your extended system.

3. **Clothing awareness:** Shift your attention to where clothing touches your skin. You had probably stopped noticing this. The body had predicted it and cancelled the signal. Now that you attend, the boundary comes alive again.

Write your response here

(For each exercise, describe what you felt. Where did your boundary expand, sharpen, or shift?)

Part 4: Two Systems Meeting (5 minutes)

If you have a partner: Stand or sit facing each other at arm's length. Both close your eyes. Slowly extend your hands toward each other until your fingertips are close but not touching. Notice: Can you feel the other person's presence before contact? What happens at the moment of touch? Two Markov blankets have met.

If alone: Bring your own two palms close together without touching. Notice the space between them. Can you feel warmth, tingling, or pressure? Slowly bring them together. Notice the moment of contact.

Write your response here

(Describe what you sensed in the space between -- before, during, and after contact.)

Part 5: Reflection

Return to stillness. Take three breaths. Consider what you have experienced.

Write your response here

(How did this lab change your understanding of what a "system" is? What did you learn about boundaries that you could not learn from a textbook?)

Reflection Table

Aspect of Systems	What I Felt	Active Inference Connection
Boundary (skin)	_____	Markov blanket separating internal from external states
Inside (interoception)	_____	Internal states of the generative model
Outside (exteroception)	_____	External states and sensory observations
Boundary disruption	_____	Prediction error at the system boundary
Boundary extension	_____	Extended cognition and tool use